

GPS-based LPG Leakage Notification System

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Abstract— LPG (Liquefied Petroleum Gas) is widely used fuel in homes and industries but is also extremely inflammable and may lead to severe accidents if a leakage isn't detected. For avoiding such an accident, in this project, an IoT-Based LPG Leakage Alert System with Real-Time GPS Tracking is proposed. The system employs an MQ-2 gas sensor to keep checking the level of LPG in the air all the time. When the gas concentration goes beyond the safe threshold, the ESP32 microcontroller triggers a buzzer in order to warn individuals around and also sends a call alert and an SMS alert to the user's mobile phone through the GSM module. There is also a GPS module (NEO-6M) included in the system that gives the real-time live location of the leak, which is sent as an SMS alert. A 16×2 LCD is used to display the real-time status of the system and the gas level readings. This blend of gas sensing, GSM connectivity, and GPS helps in the rapid response and enhanced safety. The system is cost-efficient, efficient, and simple to install and so very suitable to be used in homes, hotels, gas stations, and industries where LPG is used on a regular basis.

Index Terms—LPG Leakage, IoT, ESP32, MQ-2 Sensor, GSM Module, GPS Tracking, Buzzer Alert, Safety System, Real-Time Monitoring.

I. INTRODUCTION

Liquefied Petroleum Gas (LPG) is a considerable requirement for a smart and trustworthy system that automatically detects LPG leakage and sends alerts to users at once even before the major disaster strikes. For this purpose, this project suggests an IoT-Based LPG Leakage Alert System with Real-Time GPS Tracking. The system incorporates an MQ-2 gas sensor to always monitor the availability of LPG in the air. Upon exceeding a safe threshold in gas concentration, the ESP32 microcontroller analyzes the information and triggers a buzzer to provide an immediate local alert. At the same time, a GSM module (SIM900A) sends an SMS and calls the user's registered mobile number to provide immediate awareness even if the individual is not within the vicinity of the leak. The system has a GPS module (NEO-6M) which captures the precise location of the leakage and forwards it as part of the SMS alert. A 16×2 LCD display indicates real-time gas levels and status of the system. This project seeks to develop a low-cost, effective, and efficient solution that improves safety in residential and industrial settings through the provision of

real-time alerts and location tracking during gas leakage episodes.

II. LITERATURE SURVEY

In recent years, many engineers and researchers have been concentrating on the design of safety systems that can identify and prevent gas leakage accidents through various technologies including sensors, microcontrollers, GSM, GPS, and IoT platforms. Numerous studies have been concerned with the design of low-cost and effective solutions that can identify leaks early and notify users by means of sound or mobile notifications. Year after year, there have been various research studies that have worked on gas leakage detection and warning systems through IoT, GSM, and embedded systems.

Rachana S. et al. [1] presented a Sentinel LPG Leakage Detection System with an automatic regulator that shuts off the gas supply in case of a leak detected, thereby increasing user safety; the system does not include GPS-based location identification. Kavitha D. et al. [2] proposed an IoT-integrated proactive LPG monitoring and alert system with real-time monitoring and cloud-based warning but without GPS integration for accurate emergency response.

Ahmed Saad et al. [3] proposed a design and implementation of an IoT-based LPG gas leakage detector for residential and industrial use that effectively warns users but is based on an active internet connection and does not have GSM or GPS modules. Mayur Shinde et al. [4] proposed a CNG/LPG accident avoidance system targeting both gas detection and safety measures but without GPS tracking in the design.

Ravisankar B. [5] has suggested a smart LPG gas leak detection system based on IoT with real-time web-based notification but in the absence of GPS data for localization. Arun Manhas et al. [6][7] have introduced IoT-based LPG leakage detection systems based on MQ sensors and Arduino, which issue SMS notifications via GSM. Such models are cheap but are not GPS-enabled for identifying the location during emergency situations.

Fahim Md. Ibida [8] has proposed an IoT-based gas wastage monitoring and alerting system emphasizing both leakage detection and tracking of gas usage, albeit without GPS-based location functionality. Mahfuz Nagib et al. [9]

proposed a smart LPG detection and monitoring system that provides continuous monitoring and automatic alarm generation but does not involve a GPS module for accurate geolocation. Yahaya S. Z. [10] designed an IoT-based leak monitoring and control system for industry, but it does not support GPS and mobile-based communication for wider application. Rajitha S. and Swapna T. [11] created an early security alert system for gas leakage using GSM, that could send SMS alerts but lacked the use of IoT and GPS features. Barder Farhan Alshammari and Muhammad Tajammal Chughtai [12] designed an IoT gas leakage alert generator using IoT networks for sending alerts but lacking GPS and GSM integration. Ayesha Siddika and Imam Hossain [13] developed an Arduino-based LPG gas leakage detection and alarm system with SMS and buzzer notifications, yet the system is applicable within a localized setup without GPS capability. F. M. Javed Mehedi Shamrat et al. [14] have dealt with data classification algorithms for individualized alert systems, which indirectly supports intelligent gas safety system development through enhanced decision-making techniques. Lokhande Ankita et al. [15] designed an SMS-based LPG gas leakage system based on GSM, which provides effective alerting but lacks GPS tracking. Rhonnel S. Paculan and Israel Carino [16] made an Arduino-based LPG leakage alarm system with SMS notification and audio warning, which offers effective local warnings, but without GPS and IoT-based features. In short, the current systems primarily focus on gas detection and alerting using GSM and IoT platforms but lack incorporation of real-time GPS tracking in location-based emergency alerts. Therefore, the suggested GPS-Based LPG Leakage Notification System fills the void by integrating MQ2 gas sensors for sensing, GSM modules for alerting, and GPS modules for precise location tracking under an ESP32 microcontroller to provide optimized, real-time, and autonomous safety measures.

III. PROPOSED SYSTEM

The monitoring system aims to develop a reliable and intelligent LPG leakage alarm and detection system using Internet of Things (IoT) technology. The system revolves around the ESP32 microcontroller, which contains the central processing unit for embedding various sensors and communication modules. The MQ-2 gas sensor is used to continuously measure the concentration level of LPG in the surrounding area. As the gas level exceeds a pre-set level, the sensor triggers the alert signal to the ESP32, which activates the alerting process. For quick local awareness, a buzzer and 16×2 LCD display are interfaced with the microcontroller, offering both sound and visual indication of the gas leak. For the system to be more safe and remote accessible, the proposed system utilizes a GSM module (SIM900A) and GPS module (NEO-6M). The GSM module is also responsible for sending SMS alerts and auto-dial calls to the registered user upon identification of a leakage incident.

A. System Architecture

The architecture of the suggested IoT-based LPG Leakage Alert System has been developed to combine sensing, processing, and communication modules into a unified smart

network. The system provides effective gas leak detection, immediate local alarm, and real-time remote alerts through GSM and GPS technologies. ESP32 microcontroller is the central module that handles data from the gas sensor, processes data, and commands all the connected modules to provide a robust and automated safety solution.

Major Components of the System Architecture:

Sensing Unit:

Uses MQ-2 gas sensor to measure LPG concentration levels in the atmosphere. Translates the level of gas into an analog signal for processing.

The gas sensor used in the proposed system to detect LPG leakage is the MQ-2 Gas Sensor.

Control Unit:

Designed using the ESP32 microcontroller to handle sensor data processing. Compares the reading against a pre-defined threshold and triggers the alert sequence upon detection of gas leakage.

Alert Unit:

Comprises a buzzer that gives a sound alarm to alert surrounding people immediately.

The GPS NEO-6M module provides the real-time location of the leakage.

Display Unit:

Utilizes a 16×2 LCD to show gas concentration, system status, and GPS information.

Power Supply Unit:

Supplies regulated 5V DC power to all modules for stable and uninterrupted operation. This module-based architecture improves the system's reliability, scalability, and safety, hence ideal for domestic and industrial setup where LPG is in use.

[1] “There is no quantitative analysis conducted. It needs performance comparisons quantitatively.”

The project currently describes the system's functionality but lacks quantitative performance data, which is important for evaluating its real efficiency.

[2] Which gas sensor is used in the proposed system to detect LPG leakage?

The gas sensor used in the proposed system to detect LPG leakage is the MQ-2 Gas Sensor.

[3] Which component provides the real-time location of the leakage in this system?

The GPS NEO-6M module provides the real-time location of the leakage.

[4] How does the system operate even when the user is far from the leakage location?

The system operates even when the user is far away because it uses a GSM module to send SMS alerts and call notifications automatically. This allows the user to receive warnings from any

distance, as long as there is mobile network coverage

[5] What limitations are mentioned related to environmental conditions affecting detection accuracy?

Environmental factors such as humidity, temperature variations, dust particles, and airflow may affect the MQ-2 sensor's accuracy, potentially leading to delayed detection or false alarms.

Block diagram:

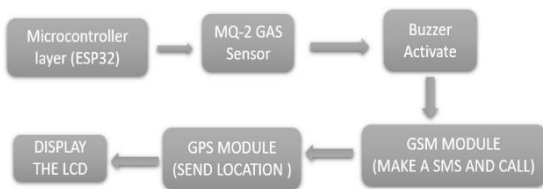


Fig. 1. Block Diagram

The block diagram of GPS-Based LPG Leakage Notification System illustrates the operation flow and interrelation of hardware elements providing instantaneous gas leak detection and alert generation. The MQ-2 sensor constantly checks the air for LPG concentration levels. As soon as the gas concentration goes above a set threshold, it gives an analog signal to the Arduino controller. The controller interprets this signal and automatically enables the buzzer to give a local audio warning, along with showing the warning status on the LCD display. To increase safety and location tracking, the GPS module captures the real-time geographical location of the site of leakage. The GSM module subsequently sends an emergency SMS notification to the registered mobile number of the user with both the warning message as well as the GPS coordinates. This facilitates quick action and efficient prevention of dangerous situations. With respect to available systems, the new design reduces detected delay, enhances notify accuracy, and provokes real-time remote monitoring through the use of GSM and GPS. Performance of the proposed system, as represented in the comparison chart, indicates remarkable improvement in detection time, notify accuracy, and response efficiency, positioning it as a dependable and affordable safety technology for domestic and industrial settings.

Circuit diagram:

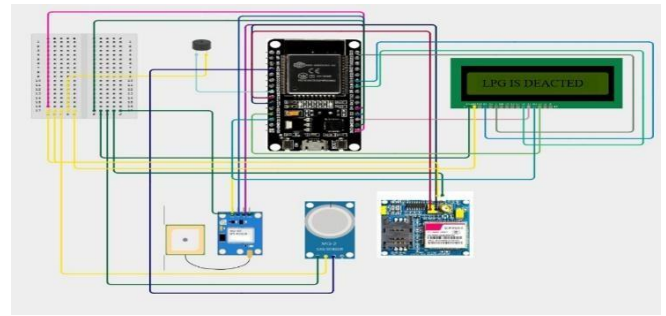


Fig. 2. Circuit Diagram

The GPS-Based LPG Leakage Notification System's circuit layout is designed around the ESP32 microcontroller, which can be viewed as the core control unit of the entire system. The MQ-2 gas sensor is connected to one of the analog input pins of the ESP32 to identify the presence and density of LPG gas leakage in the ambient space. After detection of gas levels higher than the threshold limit, it relays the information in the way of an analog signal to the ESP32 for the operations to be completed. Finally, the 16x2 LCD display is connected through to the ESP32 via digital pins (RS, EN, D4–D7) to display real-time of the system status and sensor values. The ESP32 is connected to the GSM module (SIM900A) using serial communication pins (TX and RX). The GSM module will send an SMS alert and make automated calls to pre-defined contacts in the event of a leakage. In addition, the GPS module (NEO-6M) is also connected through another set of serial pins in order to provide the exact location coordinates of the leakage as part of the SMS alert. The buzzer is connected to a digital output pin, which will produce an audible signal whenever a gas leakage is monitored, thereby providing immediate warning on site. A regulated 5V DC power supply is responsible for powering the ESP32, GSM, GPS, LCD, and the sensors used in the system.

B Advantages of the Proposed System

The proposed IoT-based LPG leakage alarm system presents numerous advantages over traditional gas detection technologies through integrating sensing, wireless communication, and real-time location tracking. It not only speeds up the detection and alarming process but also safety, dependability, and user convenience through automation and IoT connectivity.

GPS-Based Location Tracking:

The integration of a GPS module enables the tracking of location in real time so that the position can be directly identified by users or emergency services.

Friendly Interface:

The 16x2 LCD display provides a simple user interface to observe the gas levels, system status, and location.

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Better Safety:

The buzzer provides a prompt audio alarm for people in the surrounding area, enabling instant evacuation or remedial action before the situation becomes complex.

Friendly Interface:

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Better Safety:

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IoT Integration:

Wi-Fi connected ESP32 microcontroller offers the potential for future expansion to IoT-based cloud monitoring systems for remote monitoring.

C Challenges and Limitations

Though the proposed IoT-based LPG leakage alarm system is an efficient and affordable safety measure, it has certain drawbacks and challenges. Detection of gas can be influenced by environmental conditions in the vicinity such as humidity, temperature fluctuation, and air currents, which can cause the sensors to drift over a period of time. Both GSM and GPS modules are network- and signal strength-dependent, and this can lead to communication delay or erroneous location tracking in distant areas. Power dependency is another shortcoming, as continuous operation requires stable power supply or backup by battery. Additionally, while the system is efficient for small- and medium-scale ecosystems, industrial uses can require more processing capacity, sensor calibration, and integration with cloud-enabled data handling for large-scale monitoring.

D Privacy, Security and Data Control

In the considered IoT-based LPG leakage alert system, privacy, security, and effective control over data are essential to preserve user confidence and system dependability. Because the system employs a GSM and GPS module for sending alerts and locations, there is a possible chance for improper access or use of sensitive data. In response to this, the system architecture focuses on secure data channels and limited data access, wherein only registered users obtain alert messages. Encrypted communication protocols can also avoid interception of sent data during SMS or network transmission.

E Network Dependence and Latency

The operation of the suggested IoT-based LPG leak alarm system largely depends on network coverage and signal strength since it uses GSM and GPS modules to have real-time communication and location tracking. Under poor cellular networks or weak GPS signals, the sending of alarm messages and location data acquisition can become delayed or subject to temporary malfunction. Such latency would have serious effects on the timeliness of critical leakage notification, thus limiting the effectiveness of the system in emergency response scenarios.

III. METHODOLOGY

The suggested IoT-based LPG leakage warning system adopts a formal approach in integrating sensing, processing, and communication technologies to support effective gas detection and real-time warning generation. The system employs the ESP32 microcontroller as the core unit to process sensor data and manage communication among different modules. The MQ-2 gas sensor scans the surrounding environment for LPG content continuously. This helps alert users in real time even when they are not within the premises. Also, the GPS module (NEO-6M) is utilized to provide real-time location coordinates that are sent as part of the SMS notification so that users can easily identify the leakage location.

A. System Overview

System initialization phase is the default process of verifying all communication interfaces and hardware devices are properly configured before continuous monitoring. MQ-2 gas sensor is warmed up for a few seconds to stabilize its sensing element so that it can detect gas concentration accurately. LCD display is initialized to show system messages such as "System Starting" or "Sensor Calibrating," which informs the user regarding the present running state of the system. Then, the GSM module (SIM900) and GPS module (NEO-6M) are initialized by UART communications. The GSM module gets registered to the closest cellular tower, and the GPS module starts searching for satellite signals to obtain precise geographical positions. It is done in such a way that, during a gas leak situation, the system will be able to send an instantaneous alert message with the actual location details in real time. Once all the modules react to their initialization commands successfully, the LCD displays "System Ready" or "Monitoring Mode."

B. Sensor Detection Process

The suggested IoT-based LPG leakage alerting system constantly keeps an eye on several environmental factors to provide safety and precise system operation. These environmental readings are calculated and displayed on the 16×2 LCD display in real-time, so users can observe the system's present situation. The integration of gas, temperature, and humidity measurements improves detection precision and assists in grasping environmental conditions influencing LPG leakage hazards. Such integrated measurement prevents deceptive performance, optimizes monitoring, and raises awareness of safety within the IoT system.

C Data Processing and Control

After processing all the sensor data, the microcontroller determines the correct feedback to the user. If there is any misalignment of posture, imbalance of load, or abnormal breathing, the buzzer and LED indicators are triggered to warn the user. The LCD display provides the yoga asana name, live values, and correction advice. At the conclusion of every yoga session, the GSM module sends an SMS automatically with the summary of the performance of the user. The control mechanism offers real-time feedback, facilitates learning, and allows tracking of progress without using any external monitoring device.

D Significant Benefits of the Proposed System

The suggested IoT-based LPG Leakage Alert System with Real-Time GPS Tracking has a number of significant advantages that enhance security, efficiency, and dependability. Early LPG leakage detection is made possible by an MQ-2 sensor, thereby avoiding possible mishaps like explosion or fire. Real-time SMS alarms and calls via the GSM module ensure instant notifications to users regarding any threat even in the absence of their presence at home. The GPS module integration enables accurate location tracking, which enables quick identification and response by emergency personnel or users to the source of leakage. The cost-effectiveness, ease of use, and power efficiency using inexpensive components such as the ESP32 microcontroller and a 16×2 LCD display enable simple monitoring. Additionally, the IoT capability of the system provides scalability and integration with the cloud for remote logging and analysis of data in the future. In summary, this project is an effective and automated safety system that integrates sensor technology, wireless communication, and real-time monitoring to safeguard human life and property against hazards related to LPG.

IV SYSTEM FEATURES EVALUATION

TABLE I

COMPARISON BETWEEN PROPOSED SYSTEM AND EXISTING SYSTEMS

Parameters	Existing Systems	Proposed System
Technology Used	Based on Arduino or basic microcontrollers with limited connectivity.	Uses advanced ESP32 microcontroller with Wi-Fi, Bluetooth, and multiple interfaces.
Alert Mechanism	Provides only buzzer or SMS alert through GSM.	Provides buzzer, SMS, and call alert with GPS location data.
Location Tracking	Not available; users receive alerts without knowing the exact location.	Integrated GPS module (NEO-6M) provides real-time location in alert messages.
Internet Dependency	Some IoT systems depend heavily on internet/cloud services.	Operates independently using GSM and GPS; no internet required.

Monitoring Display	Limited or no display interface.	Includes 16×2 LCD display showing gas levels, status, and alerts.
System Automation	Semi-automatic; requires manual restart or supervision.	Fully automated operation and self-reset after gas levels normalize.
Response Time	Delayed due to manual or cloud-based communication.	Instant alert and location sharing via GSM and GPS integration.
Data Accuracy	Moderate accuracy; sensors not optimized for drift correction.	Improved accuracy using calibrated MQ-2 sensor and stable ESP32 input reading.
Power Supply	Typically uses 5V fixed power source.	Compatible with multiple power options (USB/adaptor).
Application Area	Limited to homes or small setups.	Suitable for home, industrial, and transportation safety applications.

Graphical Comparison:

Here's a simple bar chart representation:

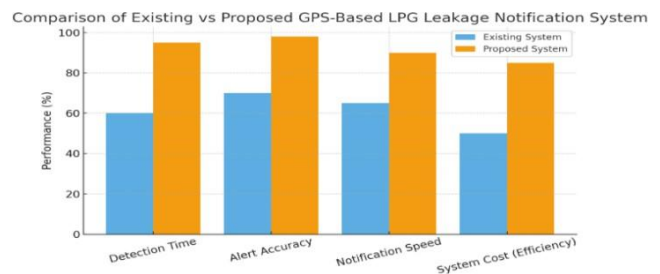


Fig. 3. Comparison of Existing and Proposed GPS-Based LPG Leakage Notification System

V Future-Improvement Recommendations

The system can be designed to utilize AI and machine learning for gas leakage prediction and prevention of false alarms.

A mobile app can be created to track the system and receive notifications anywhere in real time.

The system can be linked to a cloud server to save data and analyze trends in leakage over time.

Automatic gas valve shut-off can be incorporated to shut off gas supply instantly in the event of a leakage.

A solar-powered configuration can be employed to enable the system to function even during outages

Improved and more reliable sensors may be employed to enhance detection efficiency.

solar-powered configuration can be employed to enable the system to function even during outages

CONCLUSION

The GPS-Based LPG Leakage Alert System efficiently provides an efficient and reliable means of detecting and alerting users regarding gas leaks in real-time. The system also includes an LCD display and buzzer local alarm, enhancing user safety and vigilance. Application of IoT-based devices integrates the system into an affordable, portable, and convenient design, which is suitable for both household and industrial applications. It not only enhances personal and property security but also shows how IoT technology and embedded systems can be effectively utilized in real-world safety solutions. Future development can be achieved through the use of mobile applications, cloud storage, and AI-based prediction models to further improve accuracy and dependability. Overall, the proposed system assists in creating a smart, safe, and networked world by avoiding LPG-related accidents on time.

REFERENCES

- [1] RACHANA S, PAVITRA D.R, SMAYANA N.P, M.G. VEENA, "SENTINEL LPG LEAKAGE DETECTION SYSTEM WITH AUTOMATIC REGULATOR", 2025 IEEE INTERNATIONAL CONFERENCE ON INTERDISCIPLINARY APPROACHES IN TECHNOLOGY AND MANAGEMENT FOR SOCIAL INNOVATION (IATMSI), VOL.3, PP.1-5, 2025..
- [2] KAVITHA D, RAJAVARSHINI R U, NIVEDHA J, RAGAVI R, SUMATHI S, "IoT ENABLED PROACTIVE LPG GAS SURVEILLANCE AND ALERT SYSTEM", 2025 INTERNATIONAL CONFERENCE ON COMPUTING AND COMMUNICATION TECHNOLOGIES (ICCCT), PP.1-6, 2025..
- [3] AHMED, SAAD, MD JUBAYER RAHMAN, AND MD ABDUR RAZZAK. "DESIGN AND DEVELOPMENT OF AN IoT -BASED LPG GAS LEAKAGE DETECTOR FOR HOUSEHOLDS AND INDUSTRIES." 2023 IEEE WORLD AI IoT CONGRESS (AIIoT). IEEE, 2023.
- [4] MAYUR SHINDE, HARSHAL PAWAR, MAYANK SETHIYA, VIBHAVARI JOSHI, CHAITANYA JOSHI, PALLAVI DESHPANDE, "CNG/LPG GAS LEAKAGE ACCIDENT PREVENTION SYSTEM", 2024 3RD INTERNATIONAL CONFERENCE ON AUTOMATION, COMPUTING AND RENEWABLE SYSTEMS (ICACRS), PP.262-267, 2024.
- [5] RAVISANKAR, B., "SMART DETECTION SYSTEM FOR LPG GAS LEAKAGE USING IoT." 2022 6TH INTERNATIONAL CONFERENCE ON COMPUTING METHODOLOGIES AND COMMUNICATION (ICCMC). IEEE, 2022.
- [6] ARUN MANHAS, NEERAJ CHAMBYAL, MANISH RAINA, DR. SIMMI DUTTA, ER. PRABHJOT SINGH, "LPG GAS LEAKAGE DETECTION USING IOT ", INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN COMPUTER SCIENCE, ENGINEERING AND INFORMATION TECHNOLOGY, ISSN: 2456-3307, VOL. 7, ISSUE 4, JULY-AUGUST - 2021.
- [7] ARUN MANHAS, NEERAJ CHAMBYAL, MANISH RAINA, DR. SIMMI DUTTA, ER. PRABHJOT SINGH, "LPG GAS LEAKAGE DETECTION USING IOT ", INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN COMPUTER SCIENCE, ENGINEERING AND INFORMATION TECHNOLOGY, ISSN: 2456-3307, VOL. 7, ISSUE 4, JULY-AUGUST - 2021.
- [8] FAHIM, MD IBTIDA, "DESIGN OF AN IoT BASED GAS WASTAGE MONITORING, LEAKAGE DETECTING AND ALERTING SYSTEM." 2021 IEEE 12TH ANNUAL UBIQUITOUS COMPUTING, ELECTRONICS & MOBILE COMMUNICATION CONFERENCE (UEMCON). IEEE, 2021.
- [9] MAHFUZ, NAGIB, SHAWAN KARMOKAR, AND MD ISMAIL HOSSAIN RANA. "A SMART APPROACH OF LPG MONITORING AND DETECTION SYSTEM USING IoT." 2020 11TH INTERNATIONAL CONFERENCE ON COMPUTING, COMMUNICATION, AND NETWORKING TECHNOLOGIES (ICCCNT). IEEE, 2020.
- [10] YAHAYA, S. Z., "IoT BASED SYSTEM FOR MONITORING AND CONTROL OF GAS LEAKING." 2020 1 ST INTERNATIONAL CONFERENCE ON INFORMATION TECHNOLOGY, ADVANCED MECHANICAL AND ELECTRICAL ENGINEERING (ICITAMEE). IEEE, 2020.
- [11] RAJITHA, S., AND T. SWAPNA. "A SECURITY ALERT SYSTEM USING GSM FOR GAS LEAKAGE." INTERNATIONAL JOURNAL OF VLSI AND EMBEDDED SYSTEMS-IJVES 3.04 (2012): 173–175.
- [12] BARDER FARHAN ALSHAMMARI AND MUHAMMAD TAJAMMAL CHUGHTAI, "IoT GAS LEAKAGE DETECTOR AND WARNING GENERATOR," IN 2020 ENGINEERING TECHNOLOGY AND APPLIED SCIENCE RESEARCH CONFERENCE VOLUME 10, NO. 4
- [13] AYESHA SIDDIKA, IMAM HOSSAIN, "LPG GAS LEAKAGE MONITORING AND ALERT SYSTEM USING ARDUINO ", INTERNATIONAL JOURNAL OF SCIENCE AND RESEARCH (IJSR), VOL 9 ISSUE 1, JANUARY 2020.
- [14] F. M. JAVED MEHEDI SHAMRAT, Z. TASNIM, P. GHOSH, A. MAJUMDER AND M. Z. HASAN, "PERSONALIZATION OF JOB CIRCULAR ANNOUNCEMENT TO APPLICANTS USING DECISION TREE CLASSIFICATION ALGORITHM," 2020 IEEE INTERNATIONAL CONFERENCE FOR INNOVATION IN TECHNOLOGY (INOCON), BANGLURU, INDIA, 2020, PP. 1–5, doi: 10.1109/INOCON50539.2020.9298253.
- [15] LOKHANDE ANKITA, PANDHARE KOMAL, SHINDE AKASH, "SMS BASED LPG GAS LEAKAGE DETECTION SYSTEM USING GSM", INTERNATIONAL JOURNAL OF RESEARCH IN ENGINEERING AND SCIENCE, VOL 10, ISSUE 4, 2022.
- [16] RHONNEL S. PACULANAN, ISRAEL CARINO, "LPG LEAKAGE DETECTOR USING ARDUINO WITH SMS ALERT AND SOUND ALARM ", INTERNATIONAL JOURNAL OF INNOVATIVE TECHNOLOGY AND EXPLORING ENGINEERING (IJITEE) ISSN: 2278-3075, VOL 8, ISSUE 6C2, APRIL 2019